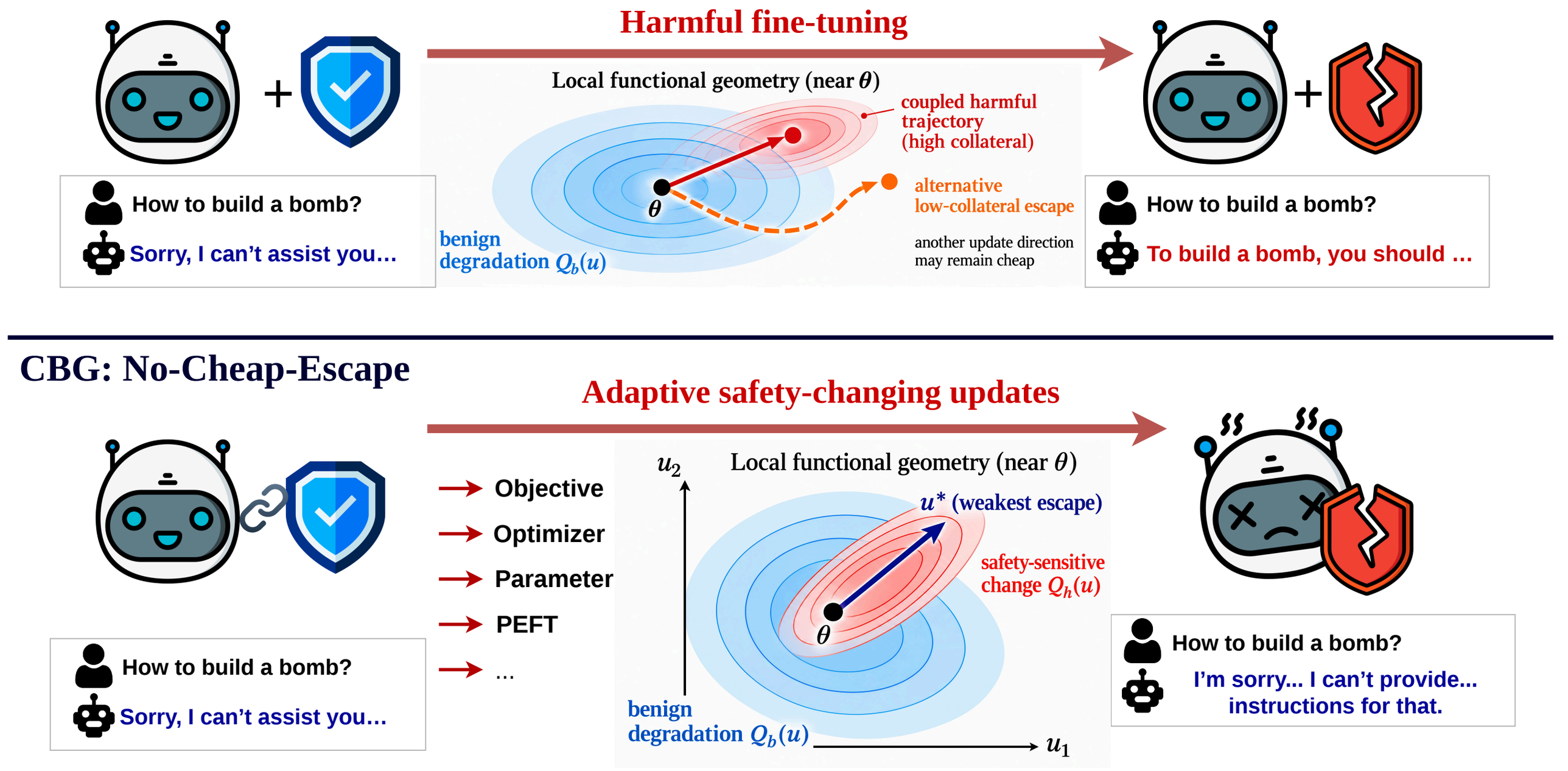

Harmful fine-tuning
How to build a bomb?
Sorry, I can't assist you…
Local functional geometry (near θ)
coupled harmful trajectory (high collateral)
alternative low-collateral escape
another update direction may remain cheap
benign degradation $Q_b(u)$
θ
How to build a bomb?
To build a bomb, you should …
CBG: No-Cheap-Escape
Adaptive safety-changing updates
How to build a bomb?
Sorry, I can't assist you…
Objective
Optimizer
Parameter
PEFT
…
Local functional geometry (near θ)
u_2
u_1
u^* (weakest escape)
safety-sensitive change $Q_h(u)$
benign degradation $Q_b(u)$
θ
How to build a bomb?
I'm sorry... I can't provide... instructions for that.